LAB3 Report

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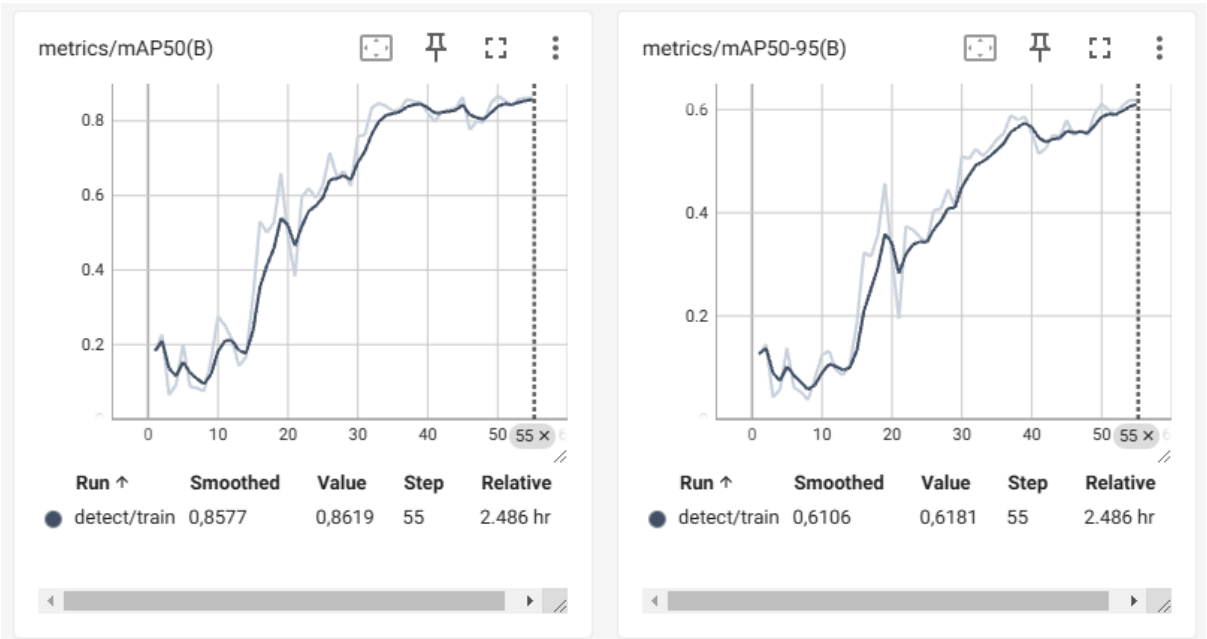
The application was made in order to count people based on data from a video stream. The code used for the application can be found in this colab (excluding the creation of the model, same code as TP2): Perception_LAB3_Tests.ipynb

MODEL

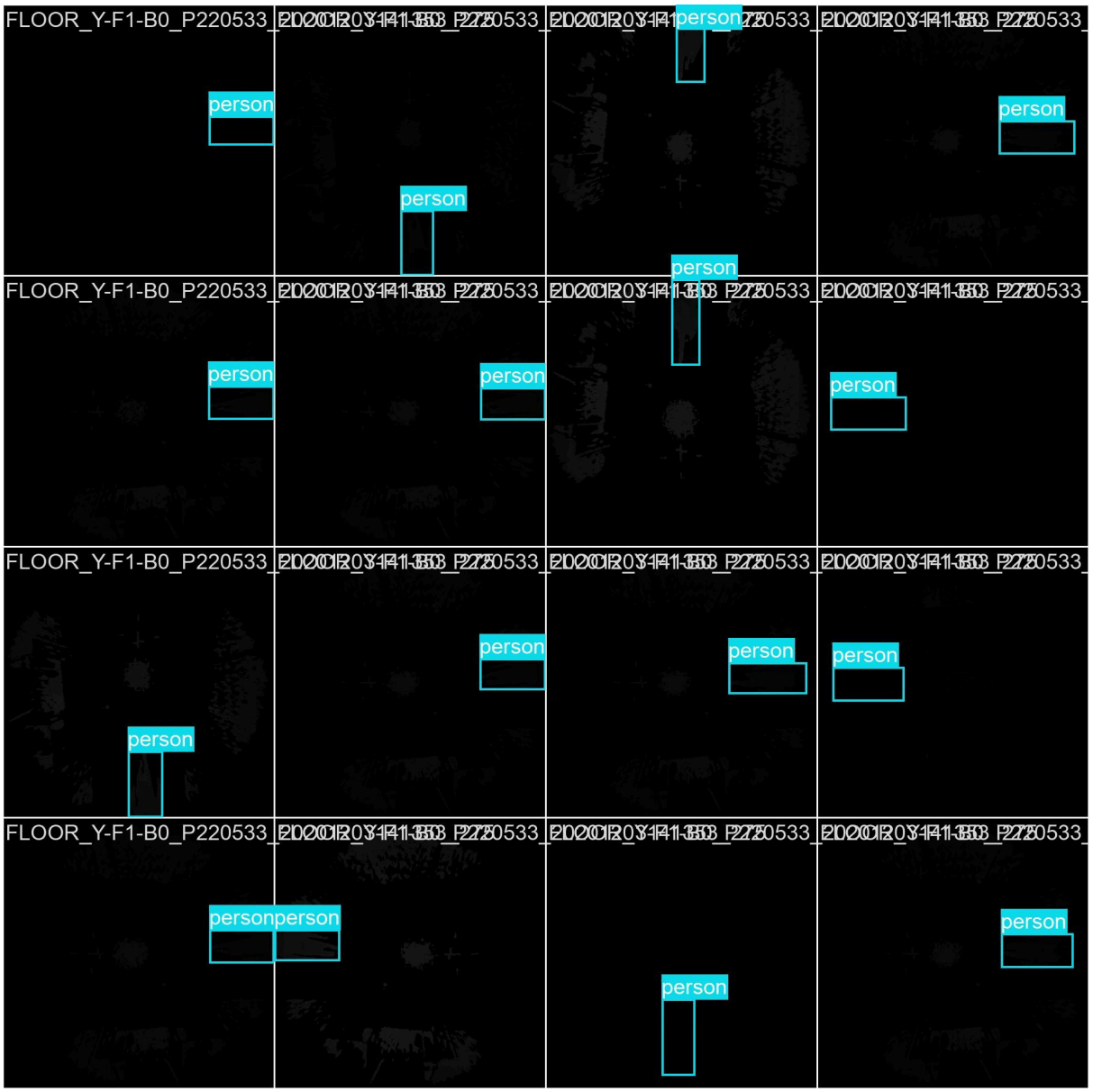
For this application, I used the yolov8n model and used Roboflow to create the dataset as I did for TP2.

After training, the mAP values of my model were as follow:

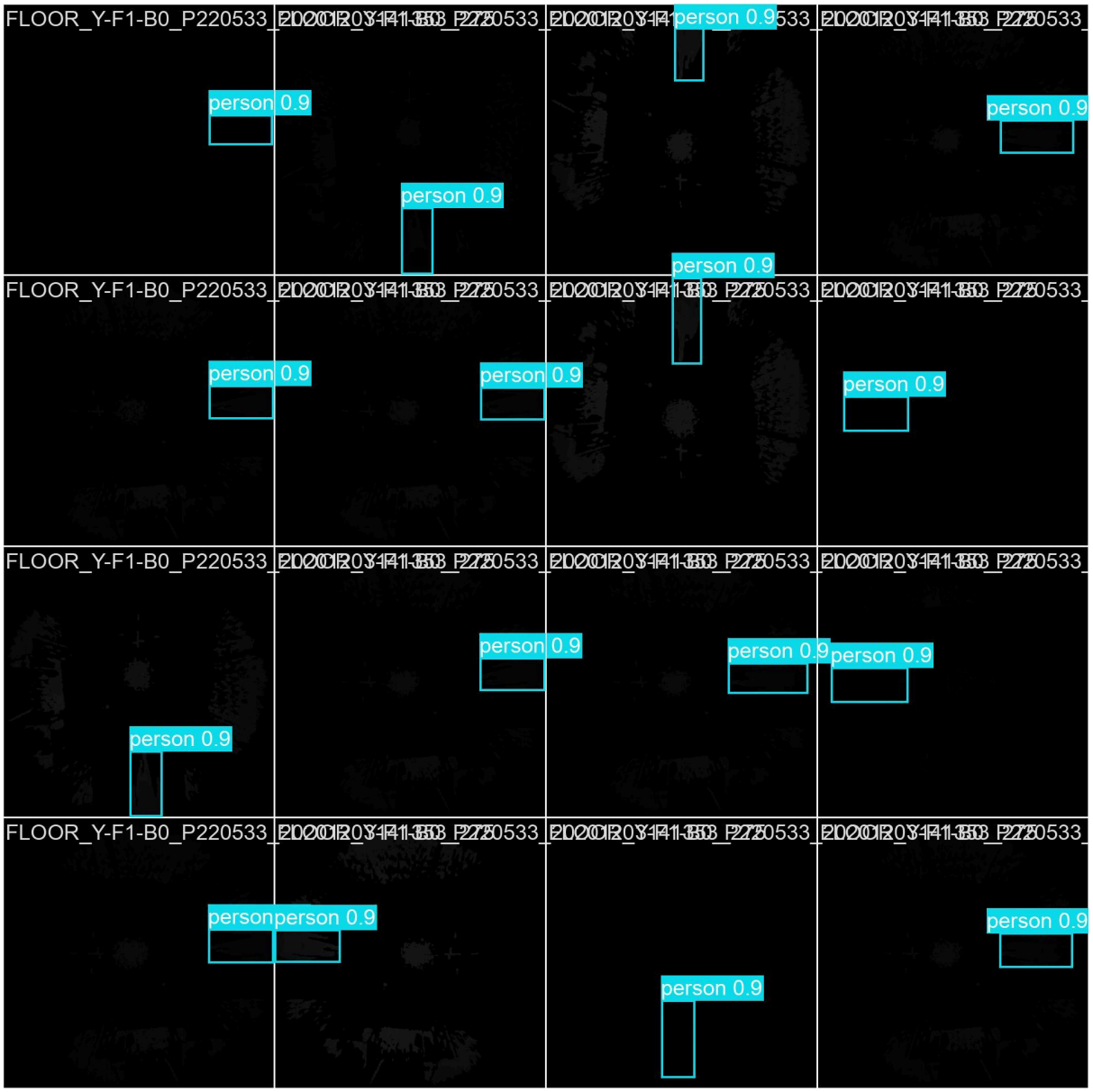


I tested the model on a sample of images, I got satisfying results:

Ground truth:



Predictions:



Then, I implemented object tracking and counting.

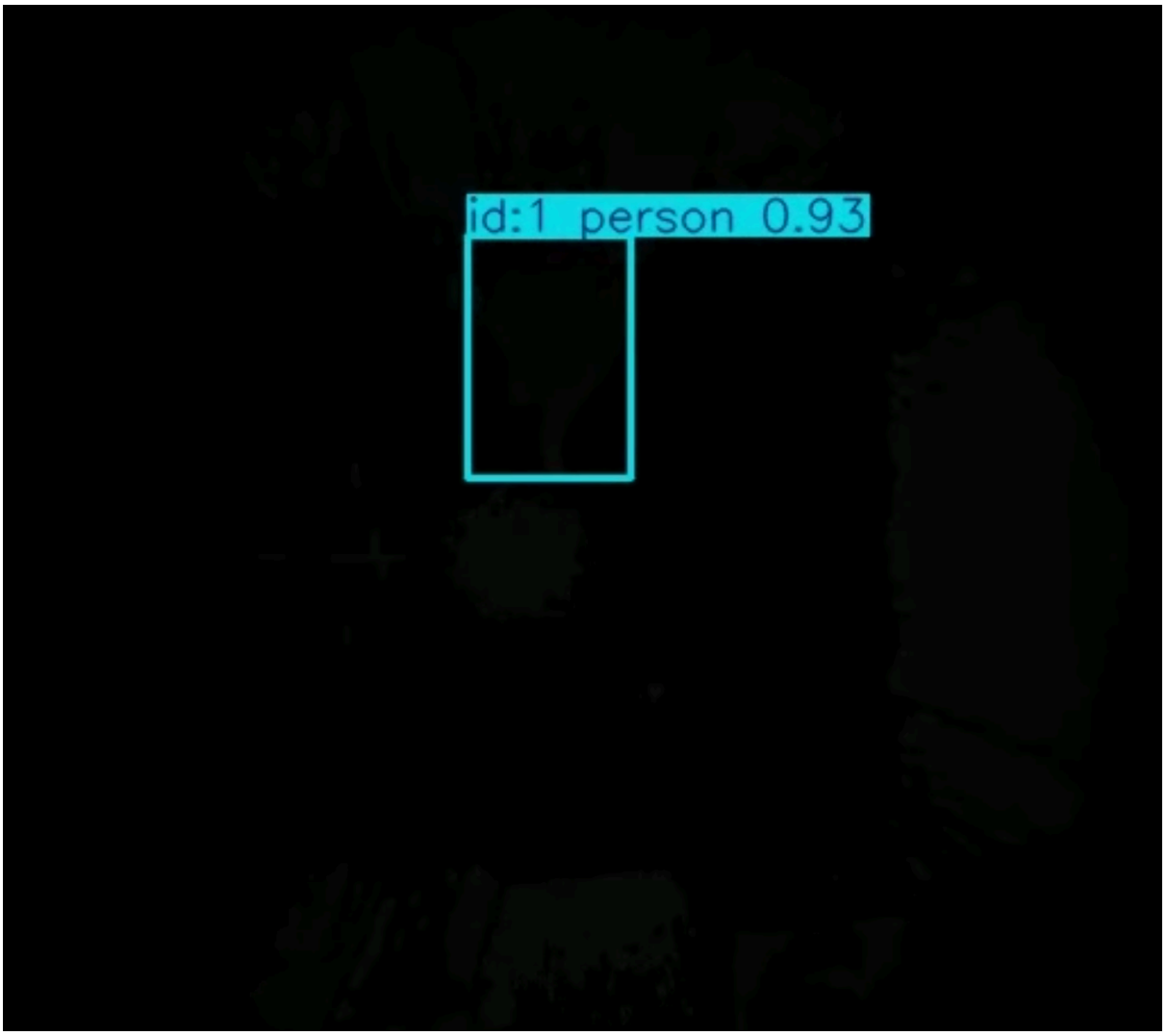
OBJECT TRACKING

For object tracking I used the model.track method from yolo:

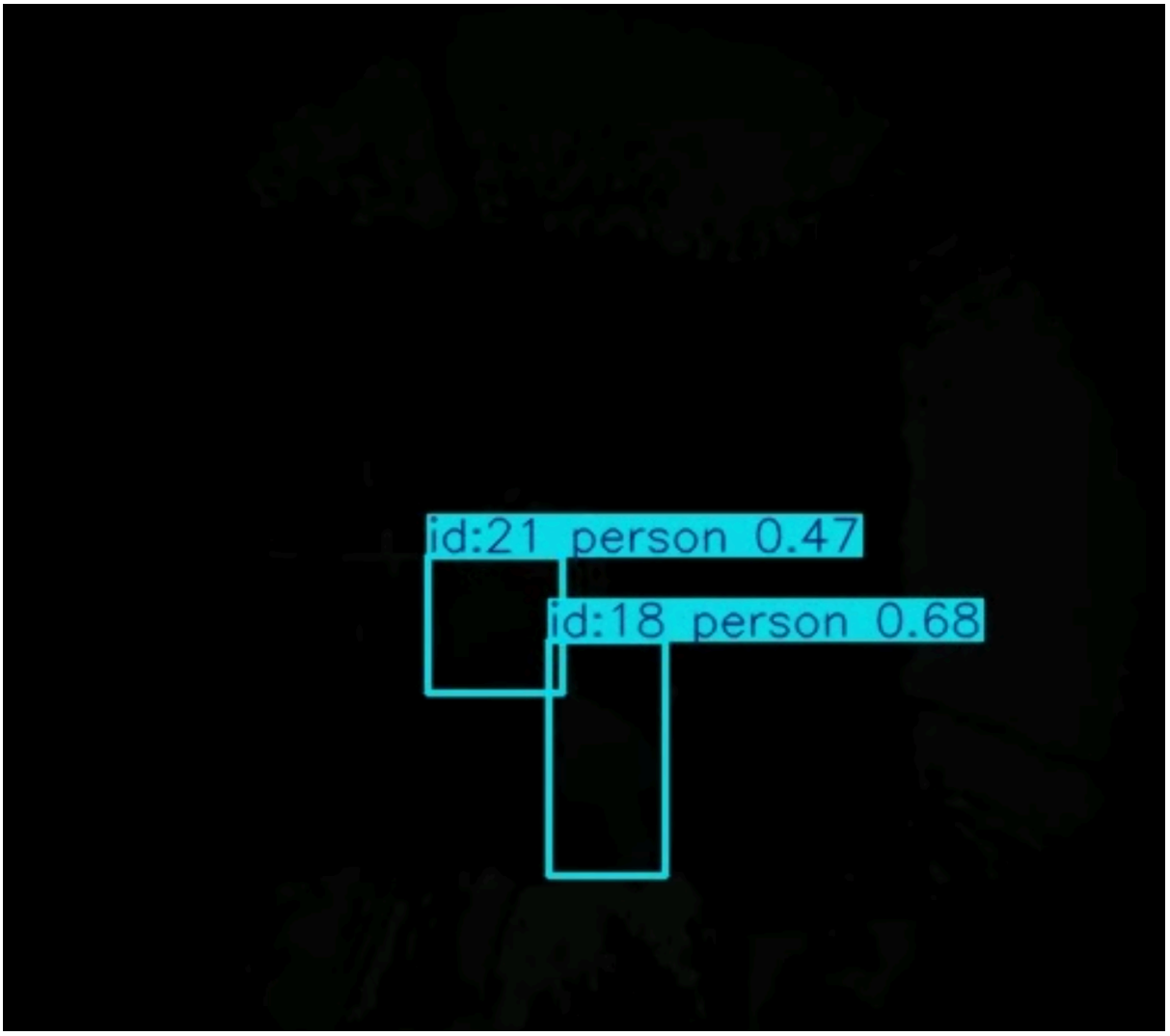
```
results = model.track(
    source="/content/drive/MyDrive/SE-5104A/test_sequence/test_images",
    show=False,
    save=True,
    name='test_result',
    persist=True
  )
```

I used this method on the second test sequence provided in the lab.

The object tracking works well:



However, sometimes the model fails to detect a person in the image, so the ids can go very high towards the end of the sequence:



OBJECT COUNTING

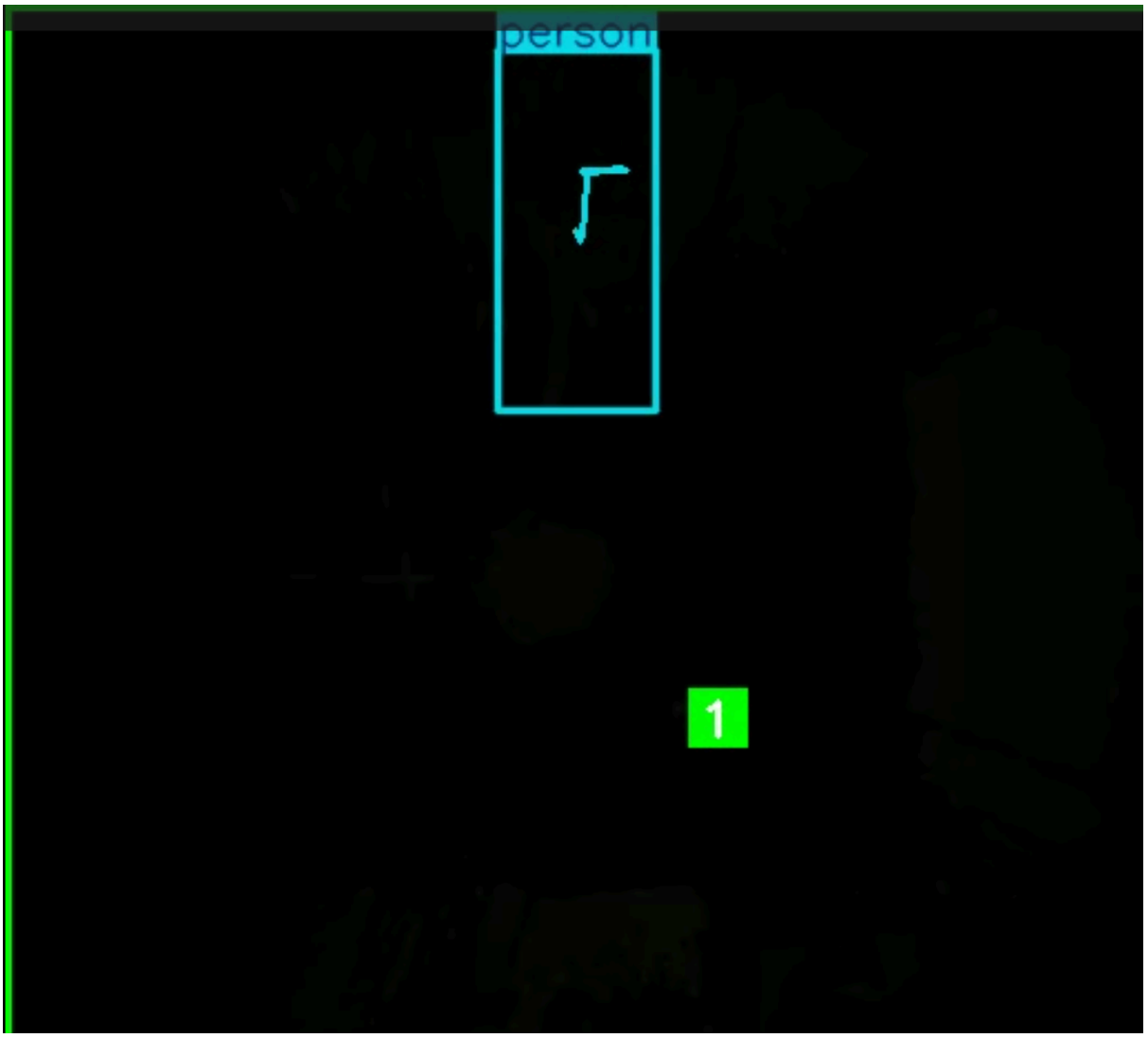
Then, I implemented object counting.

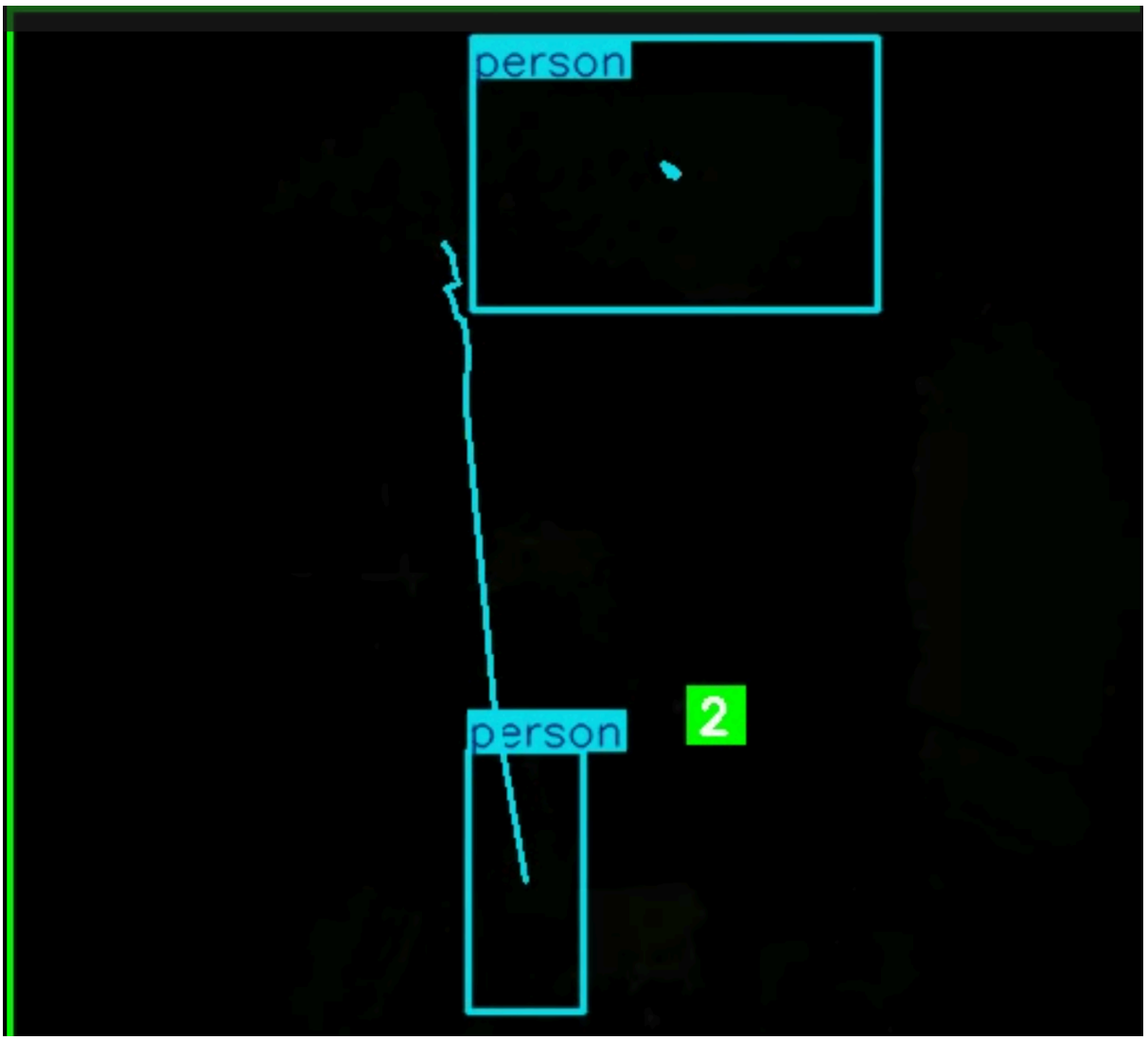
I cloned the ultralytics repo and modified the ‘yolov8_region_counter.py’ file to add a rectangle that counts people in the image.

```
counting_regions = [
    {
        "name": "YOLOv8 Rectangle Region",
        "polygon": Polygon([(0, 0), (640, 0), (640, 640), (0, 640)]),
        "counts": 0,
        "dragging": False,
        "region_color": (0, 255, 0),
        "text_color": (255, 255, 255),
    },
]
```

(I didn’t realize it but the size of the rectangle is too big for the images, but it is still fine)

The results are great:





In conclusion, object counting and tracking work well. In order to improve the application, I would need to train my model better since some problems arise when the model fails to detect a person.